Exercise 4: Worked solutions

Mathematics for Engineers I · Module 62765 · Exercise Sheet 4

From Fundamentals to AI Apocalypse · personal study guide

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Exercise 4.1: Mathematical induction

Induction starts from the first valid index and shows that each true case forces the next.

Connected topic: Induction (§2.5).

1a: Prove the finite geometric sum

Try first: check the smallest case $n = 1$, then add the next term x^{n+1} to the formula for n and put everything over $1 - x$.

Answers: see "Exercise 4.1: Mathematical induction" under Answers at the end.

1b: Prove the reciprocal-square-root bound

Try first: start the induction at $n = 2$, then show that the new term $\frac{1}{\sqrt{n+1}}$ is at least the step from $\sqrt{n}$ to $\sqrt{n+1}$.

Answers: see "Exercise 4.1: Mathematical induction" under Answers at the end.

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Exercise 4.2: Maps

Composition applies f first and g second. For finite-set counts, choose an image for each input while respecting the required property.

Connected topic: Map types and injectivity (§3.2) and counting and binomial coefficients (§2.4).

2a: Is a composition of injections injective?

Try first: assume two composite outputs are equal and unwind the definition of injectivity in reverse order.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

2b(i): Count all maps $M \to N$

Try first: each of the m elements of M independently chooses one of n target elements.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

2b(ii): Count bijections $M \to N$

Try first: A bijection requires equal cardinalities; when they match, count the possible orderings.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

2b(iii): Count injective maps $M \to N$

Try first: an injection assigns distinct targets to inputs, so there are fewer choices at each step.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

2b(iv): Count surjective maps $M \to N$

Try first: exclude maps whose image misses at least one target, correcting for maps that miss several targets.

Answers: see "Exercise 4.2: Maps" under Answers at the end.

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Exercise 4.3: Elementary functions

Addition formulas for sine and cosine imply the circle identity and double-angle relation.

Connected topic: Elementary functions and inverse functions (§3.5).

3a: Prove the circle equation

Try first: expand $\cos(x-x)$ using the cosine addition theorem.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

3b: Prove the cosine double-angle identity

Try first: use $\cos(2x)=\cos(x+x)$, then replace $\cos^2 x$ by $1-\sin^2 x$.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

3c: Evaluate cos(arcsin *x*)

Try first: let $\theta=\arcsin x$, so $\sin\theta=x$ and θ lies in the principal interval where cosine is non-negative.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

3d: Find $\cos(\frac{\pi}{5})$ exactly

Try first: use $\frac{\pi}{5}=2(\frac{\pi}{10})$ in the double-angle identity, then select the positive root because $\frac{\pi}{5}$ is in the first quadrant.

Answers: see "Exercise 4.3: Elementary functions" under Answers at the end.

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Self-Study 4.4: Maps

Vertex form reveals the range and the restriction needed for a quadratic to become one-to-one.

Connected topic: Injectivity and inverses (§3.2), composition and domains (§3.4) and elementary functions (§3.5).

4a: Classify *f*, *g*, and *h*; modify *f* and *g*

Try first: complete the square for f, solve g for its input, and inspect the coordinate relation defining h.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4b(i): Form $f \circ \mathrm{e}$

Try first: since $\mathrm{e}(x)=\mathrm{e}^x$ is always positive, substitute it into f and use the range of the downward parabola.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4b(ii): Form $g \circ f$

Try first: the inner output must satisfy $f(x)\geq 0$ because g accepts only non-negative inputs.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4b(iii): Form $g \circ \mathrm{e}$

Try first: check the exponential output against g's non-negative domain.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

4b(iv): Form $g \circ g$

Try first: the first g maps $[0,\infty)$ into $(0,1]$, which is contained in the domain $[0,\infty)$ of the second g.

Answers: see "Self-Study 4.4: Maps" under Answers at the end.

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Self-Study 4.5: Maps

Use the definitions directly: equality of composite values, existence of preimages, and the order of quantifiers.

Connected topic: Injectivity and surjectivity (§3.2).

5i: If $g \circ f$ is injective, show f is injective

Try first: assume f sends two inputs to the same value, then apply g.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

5ii: If $g \circ f$ is surjective, show g is surjective

Try first: take an arbitrary element of Z and use surjectivity of the composite to produce a preimage.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

5iii: If f and g are surjective, show $g \circ f$ is surjective

Try first: for each target z, first choose a g-preimage y, then choose an f-preimage x of y.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

5iv: Give an injective composite with non-injective g

Try first: make the domain of f a singleton, while g identifies two different middle elements.

Answers: see "Self-Study 4.5: Maps" under Answers at the end.

Self-Study 4.6: Elementary functions

Factor the quartic as a quadratic in x^2. For the parameterized quadratic, use the discriminant and include degenerate cases.

Connected topic: Polynomials and elementary functions (§3.5).

6a: Find and factor the roots

Try first: set $u{=}x^2$ for the quartic, then split the parameter cases for g_a.

Answers: see "Self-Study 4.6: Elementary functions" under Answers at the end.

6b: Show $\cos(3x)$ is not generally $3\cos x$

Try first: A single convenient counterexample disproves a universal equality; then derive the general expression.

Answers: see "Self-Study 4.6: Elementary functions" under Answers at the end.

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Self-Study 4.7: Strictly monotone functions

Strict monotonicity forbids equal outputs at distinct inputs; injectivity alone does not require order preservation.

Connected topic: Injectivity and inverse functions (§3.2).

7a: Increasing implies injective

Try first: for two distinct inputs, order them and apply the strict inequality.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

7b: Decreasing implies injective

Try first: repeat the order argument with inequalities reversed.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

7c: Is every injection strictly monotone?

Try first: find an injective rearrangement of the real line that reverses order for one pair.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

7d: Conclude that exp has an inverse

Try first: combine strict increase with the known range of the exponential.

Answers: see "Self-Study 4.7: Strictly monotone functions" under Answers at the end.

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Self-Study 4.8: Tangent addition theorems

Divide the sine and cosine addition identities only when the stated cosines are nonzero.

Connected topic: Trigonometric functions (§3.5).

8a: Prove the tangent addition and subtraction formulas

Try first: write $\tan(x \pm y) = \dfrac{\sin(x \pm y)}{\cos(x \pm y)}$, then divide numerator and denominator by $\cos x \cos y$.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

8b: Derive the tangent double-angle formula

Try first: set $y = x$ in the addition formula, observing the stated conditions keep the tangent defined.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

8c: Compute $\tan\left(\frac{\pi}{12}\right)$ exactly

Try first: apply the subtraction formula to $\dfrac{\pi}{3} - \dfrac{\pi}{4}$ and simplify by rationalizing.

Answers: see "Self-Study 4.8: Tangent addition theorems" under Answers at the end.

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Challenge 4.A: Surjective maps

Let $s(k)$ count maps onto a fixed k-element set. Partition all maps according to their image, then invert the resulting triangular relation.

Connected topics: Map types (§3.2), binomial coefficients and finite sums (§2.4) and induction (§2.5).

4.A setup: Interpret $s(k)$ and the target

Try first: relate surjectivity to the size of the image, and identify $s(n)$ with the desired count.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 1: Derive the recurrence for $s(k)$

Try first: partition the k^m maps by their exact image size j and solve for the $j{=}k$ term.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 2: Check the proposed formula for $k{=}1$, 2, and 3

Try first: use the recurrence to calculate the first values and compare them with the binomial sum.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3a: Reverse the order of a triangular sum

Try first: list the index region rather than manipulating terms.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3b: Prove the binomial coefficient identity

Try first: write both sides using factorial definitions.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 3c: Establish the convolution identity

Try first: after changing order, the coefficient of each i contains a complete alternating binomial sum.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

4.A Step 4: Prove $s(k){=}g(k)$ by induction

Try first: use $k{=}1$ as the base, then subtract the same lower-index terms from the recurrence and convolution identity.

Answers: see "Challenge 4.A: Surjective maps" under Answers at the end.

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Answers

Exercise 4.1: Mathematical induction

1a · Prove the finite geometric sum

1. Fix a real $x \neq 1$ and write S_n for the sum. The claim, for every $n \in \mathbb{N} = \{1, 2, 3, \ldots\}$, is

$$S_n = \sum_{k=0}^{n} x^k = \frac{1 - x^{n+1}}{1 - x}$$

2. **Base case $n = 1$**, the smallest natural number in this course. The sum has the two terms $1 + x$, and the right side factorises:

$$\frac{1 - x^2}{1 - x} = \frac{(1 - x)(1 + x)}{1 - x} = 1 + x = S_1$$

3. **Induction hypothesis:** the formula holds for some $n \in \mathbb{N}$. The next sum adds one term, x^{n+1}:

$$S_{n+1} = S_n + x^{n+1} = \frac{1 - x^{n+1}}{1 - x} + x^{n+1}$$

4. Put both terms over the denominator $1 - x$, which is not zero because $x \neq 1$:

$$= \frac{1 - x^{n+1} + x^{n+1} - x^{n+2}}{1 - x} = \frac{1 - x^{n+2}}{1 - x}$$

   This is the formula with $n + 1$ in place of n, so with the base case it holds for every $n \in \mathbb{N}$.

5. **Note:** the formula is also true at $n = 0$ (both sides equal 1), so starting there would prove a little more. The sheet asks for $n \in \mathbb{N}$, and this course's $\mathbb{N}$ starts at 1, so the base case is $n = 1$, as in Exercise 3.5.

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Exercise 4.1: Mathematical induction

1b · Prove the reciprocal-square-root bound

1. Write S_n for the sum. The claim starts at $n = 2$, so that is the base case: $S_2 = 1 + \frac{1}{\sqrt{2}} \approx 1.707$, while $\sqrt{2} \approx 1.414$. The claim holds at $n = 2$.

$$S_n = \sum_{k=1}^{n} \frac{1}{\sqrt{k}}$$

2. **Induction hypothesis:** assume $S_n > \sqrt{n}$ for some $n \geq 2$. Adding the next term gives

$$S_{n+1} = S_n + \frac{1}{\sqrt{n+1}} > \sqrt{n} + \frac{1}{\sqrt{n+1}}$$

3. It is enough to show that the right-hand side is at least $\sqrt{n+1}$. Multiply by the positive number $\sqrt{n+1}$:

$$\sqrt{n(n+1)} + 1 \geq n + 1 \Leftrightarrow \sqrt{n(n+1)} \geq n$$

   This holds because $n(n+1) > n^2$.

4. So $S_{n+1} > \sqrt{n+1}$: the claim at n passes to $n+1$, and with the base case it holds for every $n \geq 2$.

5. **Check:** at $n = 3$ the sum is $1 + 0.707 + 0.577 \approx 2.28$, above $\sqrt{3} \approx 1.73$. At $n = 1$ both sides equal 1, which is why the strict inequality starts at 2.

Exercise 4.2: Maps

2a · Is a composition of injections injective?

1. Suppose $(g \circ f)(x_1) = (g \circ f)(x_2)$. This says $g(f(x_1)) = g(f(x_2))$.
2. Since g is injective, $f(x_1) = f(x_2)$. Since f is injective, $x_1 = x_2$.
3. Thus $g \circ f$ is injective. The codomains need not be equal: the key is that each map is injective on the outputs that occur.

$$(g \circ f)(x_1) = (g \circ f)(x_2) \Rightarrow x_1 = x_2$$

Exercise 4.2: Maps

2b(i) · Count all maps $M \to N$

1. For each input $x \in M$, a map may assign any of the n elements of N.
2. There are n choices for each of m independent inputs, so the product rule gives $n \cdot n \cdots n = n^m$ maps.
3. If $n=0$ and $m>0$, this gives 0 maps, as expected; for $m=0$ the unique empty map is counted by $0^0=1$ under the combinatorial convention.

$$\#(M \to N) = n^m$$

Exercise 4.2: Maps

2b(ii) · Count bijections $M \to N$

1. If $m \neq n$, no bijection exists, so the count is 0.
2. If $m=n$, list the m inputs in any fixed order. The first image has m choices, the next $m-1$, and so on to 1.
3. The product is $m!$, including $0!=1$ for the empty sets.

$$\#\{\text{bijective maps } M \to N\} = m! \text{ when } m = n, \text{ otherwise } 0$$

Exercise 4.2: Maps

2b(iii) · Count injective maps $M \to N$

1. If $m>n$, the pigeonhole principle makes an injection impossible; the count is 0.
2. For $m \leq n$, order the inputs. Choose distinct images in sequence: n choices, then $n-1$, down to $n-m+1$.
3. Therefore the count is $n(n-1)\ldots(n-m+1) = n!/(n-m)!$. For $m=0$ this is 1.

$$\#\{\text{injective maps } M \to N\} = \frac{n!}{(n-m)!} \text{ when } m \leq n, \text{ otherwise } 0$$

Exercise 4.2: Maps

2b(iv) · Count surjective maps $M \to N$

1. Start with all n^m maps. For a chosen set of j target elements forbidden from the image, the remaining $n-j$ targets give $(n-j)^m$ maps.
2. There are $\mathrm{C}(n,j)$ ways to choose those j omitted targets. Inclusion–exclusion alternates the correction by the number omitted.
3. The number of onto maps is $\sum_{j=0}^{n}(-1)^j\,\mathrm{C}(n,j)(n-j)^m$. This includes the empty-set cases when interpreted with the usual combinatorial convention.

$$\#\{\text{onto maps } M \to N\} \;=\; \sum_{j=0}^{n}(-1)^j\,\mathrm{C}(n,\,j)(n\,-\,j)^m$$

Exercise 4.3: Elementary functions

3a · Prove the circle equation

1. The cosine difference formula is $\cos(a-b)=\cos a\cos b+\sin a\sin b$.
2. Set $a=b=x$. The left side is $\cos 0=1$, and the right side is $\cos^2 x+\sin^2 x$.
3. Hence $\cos^2 x+\sin^2 x=1$ for every real x.

$$\cos^2 x \;+\; \sin^2 x \;=\; 1$$

Exercise 4.3: Elementary functions

3b · Prove the cosine double-angle identity

1. The addition formula gives $\cos(x+x)=\cos^2 x-\sin^2 x$.
2. Using $\cos^2 x=1-\sin^2 x$ from the circle equation, $\cos(2x)=1-\sin^2 x-\sin^2 x=1-2\sin^2 x$.

$$\cos(2x) \;=\; 1 \;-\; 2\sin^2 x$$

Exercise 4.3: Elementary functions

3c · Evaluate cos(arcsin x)

1. By definition, $\theta=\arcsin x$ means $\sin\theta=x$, with $\theta\in[-\frac{\pi}{2},\frac{\pi}{2}]$.
2. The circle identity gives $\cos^2\theta=1-\sin^2\theta=1-x^2$. On this interval $\cos\theta\geq 0$, so take the non-negative square root.
3. Therefore $\cos(\arcsin x)=\sqrt{1\,-\,x^2}$, valid for $-1\leq x\leq 1$.

$$\cos(\arcsin\,x) \;=\; \sqrt{1\,-\,x^2},\ \text{for}\ -1\;\leq\;x\;\leq\;1$$

Exercise 4.3: Elementary functions

3d · Find $\cos(\frac{\pi}{5})$ exactly

1. The given value is $\sin(\frac{\pi}{10})=\frac{\sqrt{5}-1}{4}$. Its square is $\frac{3-\sqrt{5}}{8}$.
2. Apply $\cos(2x)=1-2\sin^2 x$ with $x=\frac{\pi}{10}$: $\cos(\frac{\pi}{5})=1-\frac{2(3-\sqrt{5})}{8}=\frac{1+\sqrt{5}}{4}$.
3. The result is positive, as expected for $\frac{\pi}{5}\in(0,\frac{\pi}{2})$.

$$\cos\left(\frac{\pi}{5}\right) = \frac{1+\sqrt{5}}{4}$$

Self-Study 4.4: Maps

4a · Classify *f*, *g*, and *h*; modify *f* and *g*

1. For $f(x)=-x^2+6x-5$, complete the square: $f(x)=-(x-3)^2+4$. Its range is $(-\infty,4]$; it is not injective on $\mathbb{R}$ because points equally spaced about 3 have equal values.
2. Restrict f to $[3,\infty)$ and use codomain $(-\infty,4]$. It is then strictly decreasing and onto that codomain. Solving $y=4-(x-3)^2$ on this branch gives $f^{-1}(y)=3+\sqrt{4-y}$, $y\le 4$.
3. For $g(x)=\frac{1}{1+x}$, $x\ge 0$, the values are $(0,1]$. It is strictly decreasing and injective, but not onto $\mathbb{R}$. With codomain $(0,1]$, it is bijective; solving $y=\frac{1}{1+x}$ gives $g^{-1}(y)=\frac{1}{y}-1$.
4. For $h(x,y)=(x,x+y,y)$, equality of outputs gives equality of the first and third coordinates, hence x and y agree: h is injective. Its image is the plane $\{(a,b,c):b=a+c\}$, so it is not onto $\mathbb{R}^3$ and is not bijective.

$$f(x) = -(x-3)^2 + 4$$

$$f^{-1}(y) = 3 + \sqrt{4-y}$$

5.

$$g^{-1}(y) = \frac{1}{y} - 1$$

Self-Study 4.4: Maps

4b(i) · Form $f\circ e$

1. Every real x is allowed: $e^x\in\mathbb{R}$, the domain of f.
2. $(f\circ e)(x)=-(e^x)^2+6e^x-5=-e^{2x}+6e^x-5$.
3. Writing $t=e^x>0$ gives $4-(t-3)^2$. Its maximum 4 occurs when $t=3$; as $t\to0^+$ or $t\to\infty$ the value tends without bound downward. The range is $(-\infty,4]$.

$$(f\circ e)(x) = -e^{2x} + 6e^x - 5, \text{ range } (-\infty, 4]$$

Self-Study 4.4: Maps

4b(ii) · Form $g\circ f$

1. Require $4-(x-3)^2\geq0$, so $(x-3)^2\leq4$ and $1\leq x\leq5$. This is the composite domain.
2. On it, $f(x)$ ranges over $[0,4]$, and $(g\circ f)(x)=\dfrac{1}{1+f(x)}=1/[5-(x-3)^2]$.
3. The denominator runs from 1 to 5; reciprocation reverses the order. The range is $[\frac{1}{5},1]$.

$$(g\circ f)(x) = \frac{1}{5-(x-3)^2}, \text{ domain } [1, 5], \text{ range } \left[\frac{1}{5}, 1\right]$$

Self-Study 4.4: Maps

4b(iii) · Form $g\circ e$

1. For all $x\in\mathbb{R}$, $e^x>0$, so the composition is defined everywhere.
2. $(g\circ e)(x)=\dfrac{1}{1+e^x}$.
3. As e^x ranges over $(0,\infty)$, this expression decreases through $(0,1)$. It approaches 1 and 0 but reaches neither, so the range is $(0,1)$.

$$(g\circ e)(x) = \frac{1}{1+e^x}, \text{ range } (0, 1)$$

Self-Study 4.4: Maps

4b(iv) · Form $g \circ g$

1. The composite domain is $[0,\infty)$, since $g(x) \geq 0$ for every allowed x.
2. $(g \circ g)(x) = 1/[1+\frac{1}{1+x}] = \frac{1+x}{2+x}$.
3. This increases from $\frac{1}{2}$ at $x=0$ and approaches 1 without reaching it. The range is $[\frac{1}{2},1)$.

$$(g \circ g)(x) = \frac{1+x}{2+x}, \text{ range } [\frac{1}{2}, 1)$$

Self-Study 4.5: Maps

5i · If $g \circ f$ is injective, show f is injective

1. Suppose $f(x_1)=f(x_2)$. Applying g gives $(g \circ f)(x_1)=(g \circ f)(x_2)$.
2. Injectivity of $g \circ f$ implies $x_1=x_2$. Thus f is injective.

$$g \circ f \text{ injective} \Rightarrow f \text{ injective}$$

Self-Study 4.5: Maps

5ii · If $g \circ f$ is surjective, show g is surjective

1. Let $z \in Z$. Since $g \circ f$ is onto Z, some $x \in X$ satisfies $g(f(x))=z$.
2. The element $y=f(x)$ belongs to Y and $g(y)=z$. Every z has a preimage under g, so g is surjective.

$$g \circ f \text{ surjective} \Rightarrow g \text{ surjective}$$

Self-Study 4.5: Maps

5iii · If f and g are surjective, show $g \circ f$ is surjective

1. Let $z \in Z$. Surjectivity of g gives $y \in Y$ with $g(y)=z$.
2. Surjectivity of f gives $x \in X$ with $f(x)=y$. Therefore $g(f(x))=z$.
3. Every z is reached by $g \circ f$, proving surjectivity.

$$f \text{ and } g \text{ surjective} \Rightarrow g \circ f \text{ surjective}$$

Self-Study 4.5: Maps

5iv · Give an injective composite with non-injective g

1. Let $X=\{a\}$, $Y=\{0,1\}$, $Z=\{z\}$. Define $f(a)=0$ and define $g(0)=g(1)=z$.
2. The map g is not injective because $0\neq 1$ but $g(0)=g(1)$.
3. The composite $X\to Z$ sends the only input a to z. A map on a singleton is injective, so $g\circ f$ is injective.

$$X = \{a\}, Y = \{0, 1\}, Z = \{z\}$$

$$f(a) = 0, g(0) = g(1) = z$$

Self-Study 4.6: Elementary functions

6a · Find and factor the roots

1. For f, divide by 5 and set $u = x^2$: $u^2 + 4u - 5 = (u - 1)(u + 5)$. Thus $f(x) = 5(x^2 - 1)(x^2 + 5)$.
2. The factor $x^2 - 1$ gives the real roots $x = \pm 1$; $x^2 + 5$ is always positive, and over $\mathbb{C}$ it has the roots $\pm i\sqrt{5}$. The complete linear factorisation is

$$f(x) = 5(x - 1)(x + 1)(x - i\sqrt{5})(x + i\sqrt{5})$$

3. For $g_a(x) = ax^2 + 2x + 1$: if $a = 0$ it is the line $2x + 1$, with the single root $-\frac{1}{2}$. If $a \neq 0$, the discriminant decides:

$$\Delta = 2^2 - 4a = 4(1 - a)$$

4. When $a < 1$ and $a \neq 0$, $\Delta > 0$ and there are two distinct real roots, so $g_a = a(x - r_+)(x - r_-)$ with

$$r_\pm = \frac{-1 \pm \sqrt{1 - a}}{a}$$

   At $a = 1$, $\Delta = 0$ and $g_a = (x + 1)^2$, a double root at -1.

5. When $a > 1$, $\Delta < 0$: there are no real roots, and over $\mathbb{C}$ the same factorisation holds with $r_\pm = \frac{-1 \pm i\sqrt{a - 1}}{a}$.
6. **Number of real roots of g_a:** two for $a < 1$ with $a \neq 0$; one for $a = 0$ (a line) and for $a = 1$ (a double root); none for $a > 1$.

Self-Study 4.6: Elementary functions

6b · Show cos(3x) is not generally 3cos x

1. Take $x=0$: $\cos(3\cdot 0)=1$ while $3\cos 0=3$, so they are not equal in general.
2. Use $\cos(2x+x)=\cos(2x)\cos x-\sin(2x)\sin x$. Substitute $\cos 2x=\cos^2 x-\sin^2 x$ and $\sin 2x=2\sin x\cos x$.
3. This gives $\cos 3x=\cos^3 x-3\sin^2 x\cos x=\cos^3 x-3(1-\cos^2 x)\cos x=4\cos^3 x-3\cos x$.

$$\cos(3x) = 4\cos^3 x - 3\cos x$$

Self-Study 4.7: Strictly monotone functions

7a · Increasing implies injective

1. Take $x\neq y$. Either $x<y$ or $y<x$. If $x<y$, strict increase gives $f(x)<f(y)$, hence $f(x)\neq f(y)$.
2. If $y<x$, strict increase gives $f(y)<f(x)$, again unequal outputs. Therefore f is injective.

$$x < y \Rightarrow f(x) < f(y) \Rightarrow f(x) \neq f(y)$$

Self-Study 4.7: Strictly monotone functions

7b · Decreasing implies injective

1. Take $x\neq y$. If $x<y$, strict decrease gives $f(x)>f(y)$; if $y<x$, it gives $f(y)>f(x)$.
2. In either case the outputs differ, so f is injective.

$$x < y \Rightarrow f(x) > f(y) \Rightarrow f(x) \neq f(y)$$

Self-Study 4.7: Strictly monotone functions

7c · Is every injection strictly monotone?

1. Define $f:\mathbb{R}\to\mathbb{R}$ by swapping 0 and 1 and leaving every other real number fixed. This is a bijection, hence injective.
2. Since $0<\frac{1}{2}$ but $f(0)=1>f(\frac{1}{2})=\frac{1}{2}$, it is not increasing. Since $\frac{1}{2}<1$ but $f(\frac{1}{2})=\frac{1}{2}>f(1)=0$, it is not decreasing.
3. Thus injectivity does not imply strict monotonicity.

$$f(0) = 1 > \frac{1}{2} = f\left(\frac{1}{2}\right)$$

$$f\left(\frac{1}{2}\right) = \frac{1}{2} > 0 = f(1)$$

Self-Study 4.7: Strictly monotone functions

7d · Conclude that exp has an inverse

1. The exponential is strictly increasing on $\mathbb{R}$, so by part (a) it is injective.
2. Its range is $(0,\infty)$, so $\exp:\mathbb{R}\to(0,\infty)$ is also surjective.
3. It is therefore bijective and has an inverse, the natural logarithm $\ln:(0,\infty)\to\mathbb{R}$.

$$\exp : \mathbb{R} \to (0,\, \infty) \text{ is bijective, } \exp^{-1} \,=\, \ln$$

Self-Study 4.8: Tangent addition theorems

8a · Prove the tangent addition and subtraction formulas

1. The addition identities give $\sin(x\pm y)=\sin x\cos y\pm\cos x\sin y$ and $\cos(x\pm y)=\cos x\cos y\mp\sin x\sin y$.
2. The hypotheses ensure $\cos x$, $\cos y$, and $\cos(x\pm y)$ are nonzero, so division is valid. Divide top and bottom by $\cos x\cos y$.
3. The result is $\tan(x\pm y)=\dfrac{\tan x\pm\tan y}{1\mp\tan x\tan y}$. For the plus case use matching upper/lower signs; for the minus case use opposite signs.

$$\tan(x \,\pm\, y) \,=\, \frac{\tan x \,\pm\, \tan y}{1 \,\mp\, \tan x\, \tan y}$$

Self-Study 4.8: Tangent addition theorems

8b · Derive the tangent double-angle formula

1. Use $\tan(x+x)=\dfrac{\tan x+\tan x}{1-\tan^2 x}$.
2. Thus $\tan(2x) \,=\, \dfrac{2\tan x}{1-\tan^2 x}$, whenever x and $2x$ avoid the excluded tangent poles.

$$\tan(2x) \,=\, \frac{2\,\tan x}{1 \,-\, \tan^2 x}$$

Self-Study 4.8: Tangent addition theorems

8c · Compute $\tan(\frac{\pi}{12})$ exactly

1. Use $\tan(\frac{\pi}{3})=\sqrt{3}$ and $\tan(\frac{\pi}{4})=1$. Then $\tan(\frac{\pi}{12})=\frac{\sqrt{3}-1}{1+\sqrt{3}}$.
2. Multiply numerator and denominator by $\sqrt{3}-1$: the denominator becomes $3-1=2$ and the numerator becomes $(\sqrt{3}-1)^2=4-2\sqrt{3}$.
3. Therefore $\tan(\frac{\pi}{12})=2-\sqrt{3}$, a positive value as expected for an angle of 15°.

$$\tan\left(\frac{\pi}{12}\right) = 2 - \sqrt{3}$$

Challenge 4.A: Surjective maps

4.A setup · Interpret $s(k)$ and the target

1. For a fixed k-element set A, $s(k)$ is the number of maps $f{:}M{\to}A$ whose image is all of A.
2. A map $M{\to}N$ is surjective exactly when its image has n elements, so $s(n)$ counts the onto maps from M to N.
3. Assume $m=|M|\geq 1$ for the stated Step 1 recurrence. If $m=0$, there is one onto map when $n=0$ and none when $n>0$; that case is separate from this recurrence.

$$f \text{ is onto} \Leftrightarrow |f(M)| = |N| = n$$

Challenge 4.A: Surjective maps

4.A Step 1 · Derive the recurrence for $s(k)$

1. There are k^m total maps $M{\to}A$. If the image has size j, first choose its j-element image B in $\mathrm{C}(k,j)$ ways, then choose a map onto B in $s(j)$ ways.
2. Summing over image sizes gives $k^m=\sum_{j=1}^{k}\mathrm{C}(k,j)s(j)$. There is no empty image because $m\geq 1$.
3. The $j=k$ term is $s(k)$, since $\mathrm{C}(k,k)=1$. Rearranging gives $s(k)=k^m-\sum_{j=1}^{k-1}\mathrm{C}(k,j)s(j)$.

$$k^m = \sum_{j=1}^{k} \mathrm{C}(k,\, j)s(j)$$

$$s(k) = k^m - \sum_{j=1}^{k-1} \mathrm{C}(k,\, j)s(j)$$

Challenge 4.A: Surjective maps

4.A Step 2 · Check the proposed formula for k=1, 2, and 3

1. For k=1, $s(1)$=1^m=1. The proposed sum is C(1,1)1^m=1.
2. For k=2, $s(2)$=2^m−2$s(1)$=2^m−2. The guess gives C(2,1)$(-1)^1 1^m$+C(2,2)2^m=2^m−2.
3. For k=3, $s(3)$=3^m−3$s(1)$−3$s(2)$=3^m−3·2^m+3. The guess gives the same alternating expression. These cases support, but do not prove, $s(k)=\sum_{j=1}^{k} \mathrm{C}(k,j)(-1)^{k-j}j^m$.

$$s(1) = 1$$

$$s(2) = 2^m - 2$$

$$s(3) = 3^m - 3 \cdot 2^m + 3$$

Challenge 4.A: Surjective maps

4.A Step 3a · Reverse the order of a triangular sum

1. The left-hand iterated sum first chooses j=1,…,k, then i=1,…,j. Its index pairs are exactly $\{(i,j):1\le i\le j\le k\}$.
2. If instead i is fixed first, i ranges from 1 to k and j ranges from i to k. Thus $\sum_{j=1}^{k}\sum_{i=1}^{j}(\ldots) = \sum_{i=1}^{k}\sum_{j=i}^{k}(\ldots)$.
3. Both sides add the same term once for every pair in that finite triangular region.

$$\sum_{j=1}^{k}\sum_{i=1}^{j}(\ldots) = \sum_{i=1}^{k}\sum_{j=i}^{k}(\ldots)$$

Challenge 4.A: Surjective maps

4.A Step 3b · Prove the binomial coefficient identity

1. For $0\le i\le j\le k$, C(k,j)C(j,i)=$k!/[j!(k-j)!]\cdot j!/[i!(j-i)!]$.
2. Cancel $j!$ to get $k!/[i!(j-i)!(k-j)!]$.
3. On the other hand, C(k,i)C($k-i$,$j-i$)=$k!/[i!(k-i)!]\cdot(k-i)!/[(j-i)!(k-j)!]$, the same expression.

$$\mathrm{C}(k,\,j)\mathrm{C}(j,\,i) \;=\; \mathrm{C}(k,\,i)\mathrm{C}(k-i,\,j-i)$$

Challenge 4.A: Surjective maps

4.A Step 3c · Establish the convolution identity

1. Start with $\sum_{j=1}^{k} \mathrm{C}(k,j)g(j)$ and insert $g(j)=\sum_{i=1}^{j} \mathrm{C}(j,i)(-1)^{j-i}\, i^m$.
2. Use Step 3a to reverse the order, then Step 3b: $\mathrm{C}(k,j)\mathrm{C}(j,i)=\mathrm{C}(k,i)\mathrm{C}(k-i,j-i)$.
3. The result is $\sum_{i=1}^{k} \mathrm{C}(k,i)i^m \sum_{j=i}^{k} \mathrm{C}(k-i,j-i)(-1)^{j-i}$. Put $r=j-i$. The inner sum is $\sum_{r=0}^{k-i} \mathrm{C}(k-i,r)(-1)^r=(1-1)^{k-i}$.
4. It is zero unless $i=k$, when it equals 1. The whole expression therefore reduces to k^m, proving $k^m=\sum_{j=1}^{k} \mathrm{C}(k,j)g(j)$.

$$k^m \;=\; \sum_{j=1}^{k} \mathrm{C}(k,\,j)g(j)$$

Challenge 4.A: Surjective maps

4.A Step 4 · Prove $s(k)=g(k)$ by induction

1. For $k=1$, $s(1)=1$ and $g(1)=\mathrm{C}(1,1)(-1)^0 1^m=1$.
2. Assume $s(j)=g(j)$ for every $1\le j<k$. The recurrence gives $s(k)=k^m-\sum_{j=1}^{k-1} \mathrm{C}(k,j)s(j)$.
3. The convolution identity says $k^m=\sum_{j=1}^{k} \mathrm{C}(k,j)g(j)$, so isolating its final term gives $g(k)=k^m-\sum_{j=1}^{k-1} \mathrm{C}(k,j)g(j)$.
4. The induction hypothesis makes these right sides equal, hence $s(k)=g(k)$. This proves the formula for every $k\ge 1$.

$$s(k) \;=\; g(k) \;=\; \sum_{j=1}^{k} \mathrm{C}(k,\,j)(-1)^{k-j} j^m$$

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